UNIVERSITÀ DEGLI STUDI DI PADOVA

Dipartimento di Fisica e Astronomia "Galileo Galilei"

Master Degree in Physics of the Fundamental Interactions

Introduction to research

Development of detectors for monitored neutrino beams

Internship supervisor
Andrea Longhin
Internship co-supervisor
Fabio Pupilli

Candidate
Davide Marchesini

Academic Year 2023/2024

Contents

1 Abstract **2**

2 Introduction **2**

3 The ENUBET Demonstrator **2**

4 Simulation of cosmic rays **5**

5 Analysis of the real data **8**

6 Uniformity **11**

7 Event reconstruction **14**

8 Conclusion **18**

1 Abstract

The Enhanced Neutrino Beams from Kaon Tagging (ENUBET [3]) project aims to demonstrate the feasibility of a monitored neutrino beam capable to provide a flux measurement error of $\approx 1\%$.

To test the instrumented decay tunnel a demonstrator has been built at Laboratori Nazionali di Legnaro composed of a iron-scintillator sampling calorimeter.

The object of this report is to study the uniformity of the detector and the event reconstruction capabilities with the use of cosmic muons. A Geant4 simulation is employed to study the differences in response for each channel. A tracking algorithm is used to determine the angular resolution of the detector.

In the end a final evaluation of all the information is reported and the conclusion provides insight in the internship, for future objects.

2 Introduction

The ENUBET project aims to demonstrate the feasibility of a monitored neutrino beam in which the flux is constrained at the 1% level. This endeavor focuses on achieving high-precision measurements of ν_e and ν_μ cross sections at the GeV scale and conducting precise searches for physics beyond the Standard Model within the three-neutrino sector. The traditional near-far cancellation technique, which is currently employed, faces intrinsic limitations, with systematic uncertainties dominating the physics reach of future long-baseline oscillation experiments (DUNE, Hyper-Kamiokande...). ENUBET is designed to alleviate these limitations by tagging the decay products of π and K in the neutrino production process.

The current experimental activities, as of July 2024, are centered around two main components: the development and testing of the ENUBET Demonstrator, detailed in this report, and the Geant4 simulations for the future experiment, particularly focusing on the decay line and the instrumented tunnel.

3 The ENUBET Demonstrator

To test the feasibility of the instrumented decay tunnel, a demonstrator has been constructed at Laboratori Nazionali di Legnaro. This section outlines the experimental setup used throughout the project.

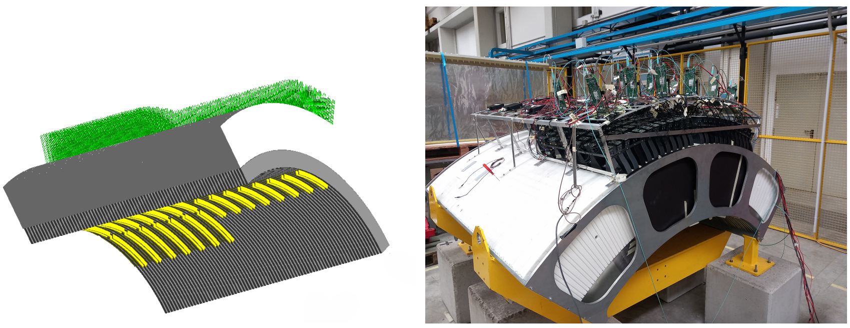


Figure 1: On the left an image of the ENUBET Demonstrator from the Geant4 simulation, in green there are the optic fibers, in yellow the t_0 (which is a layer of scintillators that will be explained later). On the right a photo with the readout boards connected.

The demonstrator consists of sampling calorimeters, where each module, called Lateral Compact Module (LCM), includes five iron tiles measuring $3 \times 3 \times 1.5$ cm^3 interleaved with five plastic scintillator tiles measuring $3 \times 3 \times 0.7$ cm^3 along the $x \times y \times z$ directions. Thus, each LCM measures $3 \times 3 \times 11$ cm^3, accounting for slight variations due to the optical fiber readout diameter and optical painting of the scintillators. The length along the z-axis is chosen so to have the equivalent length of $4.3X_0$. In the decay tunnel the kaon or electron beam would be aligned parallel to the beamline and so along z.

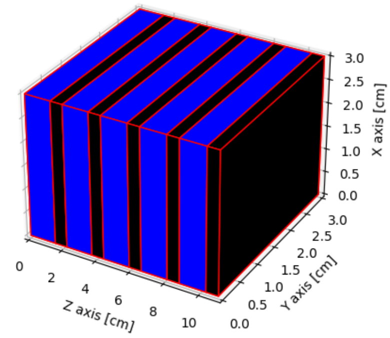


Figure 2: Drawing of a single LCM, in blue the scintillator tiles and in black the iron ones (not to scale).

The demonstrator is structured with LCM modules arranged along the z-axis, totaling 14 modules, and along the ϕ angle, comprising either 10 or 26 modules. The section with 26 modules and 10 modules are referred to as the front and back sections, respectively. It is important to note that the angular sector with $\phi = 12$ for the front part is not instrumented.

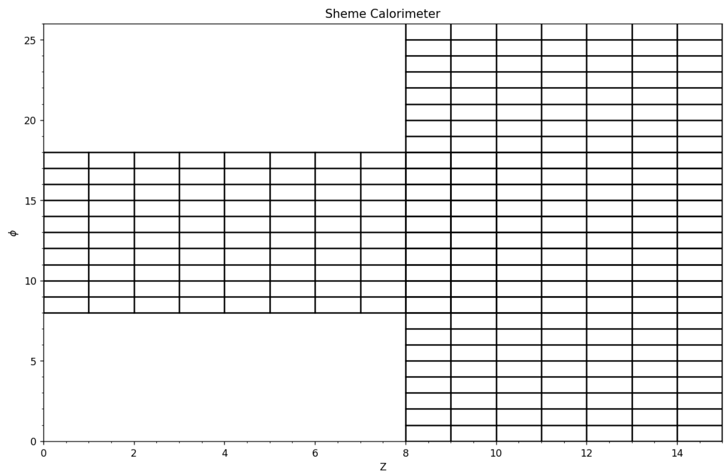


Figure 3: Scheme of a single radial layer, that it will be used through the analysis. The horizontal axis indicate the z numbering while the vertical one the ϕ numbering. Each component is then numbered by the triplet (r, ϕ, z) that identify the precise channel.

There are 3 layers such as this all with the same spatial configuration and shaped to have a perfect hermeticity and reduce dead regions, so in a trapezoidal configuration pointing toward the center. The angular coverage is from 66.6° to 113.4° for the front section and from 81° to 99° for the back one. The angular coverage of each ϕ module is 1.8°. The bottom LCM layer (which is considered the structural inner surface of the demonstrator) is at a distance from the center of 1 m.

Below the inner most LCM layer there is a so called "t_0 doublet" which is composed of two scintillators, with the same dimensions as the ones on the LCM module, acting as π^0 rejector. This pions come from hadronic decay modes of kaons and so constitute a background, not being associated to a neutrino production.

Each scintillator piece (so both for the LCM and for the t_0) incorporates two WLS fibers of 1 mm diameter inserted and glued into grooves along two opposite sides of each scintillator tile, passing through 30 cm of borated polyethylene shielding slabs for radiation protection. Each LCM is equipped then with 10 fibers, coupled to a Silicon PhotoMultiplier (SiPM) with an active area of 4×4 mm^2 and a cell size of 40 μm. Each t_0 doublet is coupled to SiPMs of 3×3 mm^2 size.

The optical fibers for each module $3LCM + 2t_0$ are housed within a black plastic "fiber concentrator", to route the fibers towards the SiPMs and to protect them from possible external damage. The SiPMs are attached on top of the concentrator and coupled via optical oil to the fibers.

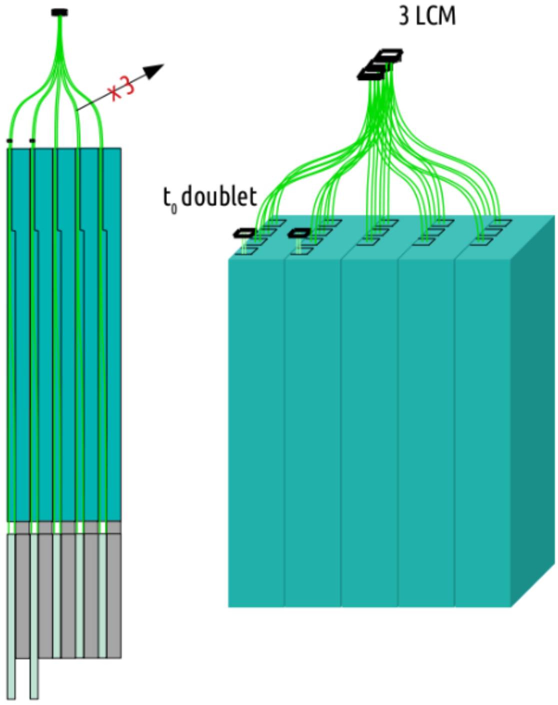


Figure 4: The scheme of a concentrator section of $3LCM + 2t_0$.

Each readout output of the SiPMs is connected to a primary PCB board, called IC board, in which there is also the bias voltage for the SiPM, which is controlled by the CAEN board.

The SiPMs' readout signals are processed by a 64 channel board CAEN A5202, which returns only the signals' amplitude and time of trigger of the signal. For the synchronization of the boards it has been used a "concentrator" named DT5215. It's working process is to synchronize all the CAEN boards via a daisy link, connected through optical fibers to each other creating a closed chain on the concentrator. This allows to work with all the available boards without losing the synchronization between each other. The concentrator is directly linked to the controlling computer, through Ethernet cable. Thanks to the Janus interface (an open source software provided by CAEN for the control and readout of FERS-5200 boards) it is possible to control all the boards simultaneously and collect the data.

Each experimental run generates a board-based .root file containing boardID, trigger timestamp, and both a low and a high amplified read-outs of each channel signal (called respectively low gain, LG, and high gain values, HG).

For each run it is possible to chose a "majority" which is the number of channels of each board that return a positive trigger. So in this way the trigger level is not managed between all the boards but for single board, it is called auto trigger mode or board-based trigger.

More information can be found in [5] and [1] for the ENUBET project and demonstrator layout.

4 Simulation of cosmic rays

Given the Geant4 simulation of the entire detector it is possible to simply generate a root file with the chosen distribution in position of the generated events, momentum direction and energy.

As a starting point, a uniform distribution was applied for a plane at 150 cm from the center as show in the figure 5.

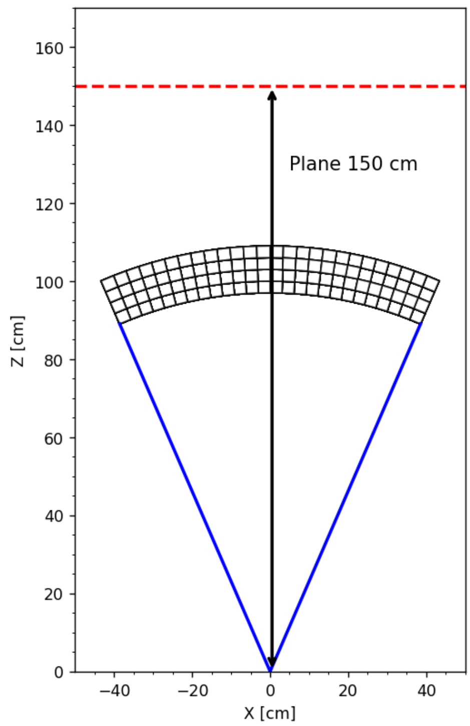


Figure 5: In red the plane considered for the event generation starting position compared to the center of the detector.

Due to the not infinite size of the plane the minimum angle elevation is around 0.04°, which is negligible. For the muon momentum direction there is a correlation between elevation angle and momentum magnitude. I assumed the formula for the spectrum of the events as

$$J(p, \theta) = f(p) \cdot (\cos\theta)^{n(p)}$$

where $f(p)$ and $n(p)$ are functions of p in GeV so that

$$f(p) = \left[1600 \cdot \left(\frac{p}{p_0} + 2.68\right)^{-3.175} \cdot \left(\frac{p}{p_0}\right)^{0.279}\right]$$
$$n(p) = \max\left[0.1, 2.856 - 0.655 \cdot \ln\left(\frac{p}{p_0}\right)\right]$$

with $p_0 = 1$ GeV/c and $p > 0.040$ GeV/c [7]. This is a good approximation and depends only on two simple parameters p and θ. Examples of the function shape are reported in figure 6 while the total energy spectrum of the final muons distribution is plotted in figure 7. As it is possible to see it is in good agreement with the PDG definition of the expected spectrum shape [4].

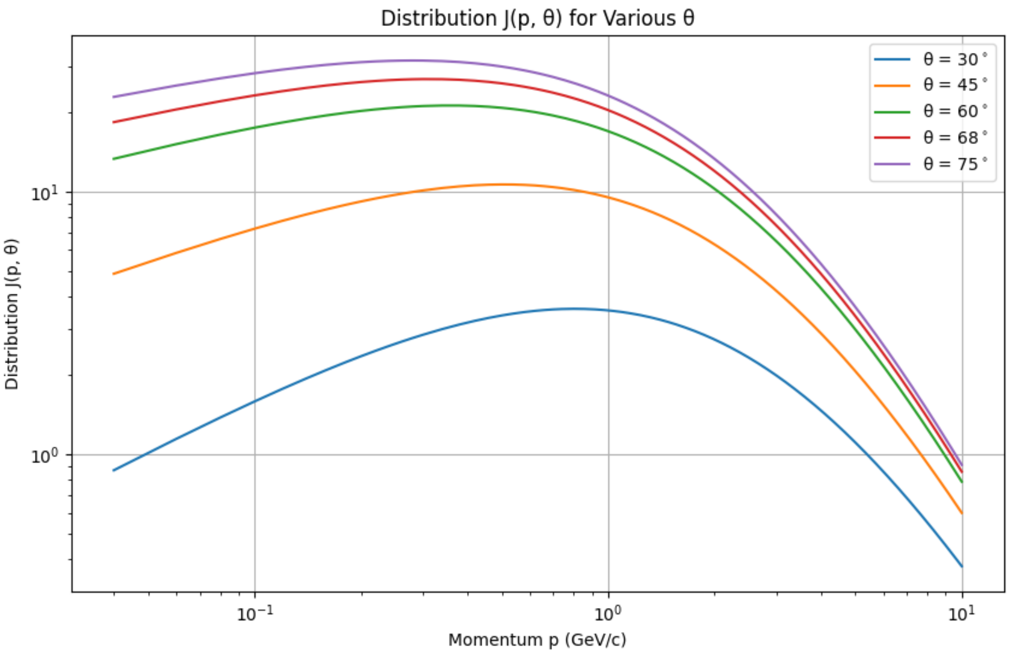


Figure 6: Examples of the $J(p, \theta)$ distribution.

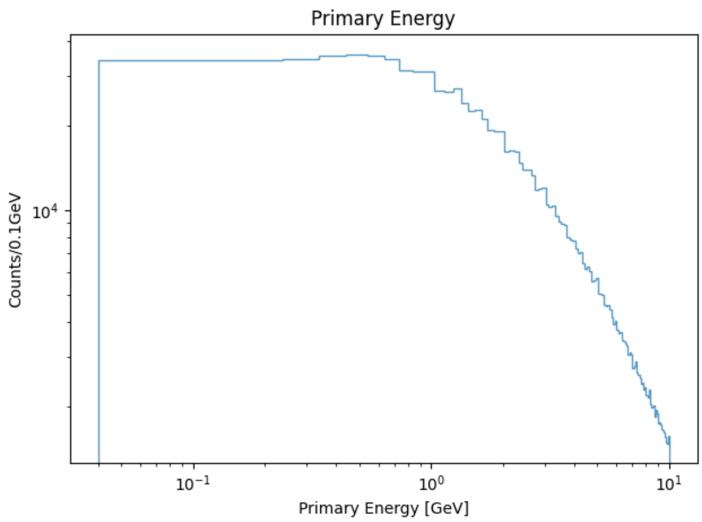


Figure 7: Total muon primary energy distribution.

With the event generated it is then possible to launch the simulation and collect the data, the sample size is 10^6 events. A first filtering has been applied to the output of the simulation in order to eliminate all the events going outside the detector and all the events entering the detector but in a region which is not read out. We have then imposed a majority equal to the one of the data run, which is $majority = 3$ and a cut on the deposited energy, $E \geq$0.5 MeV, to simulate the threshold cuts applied in the data.

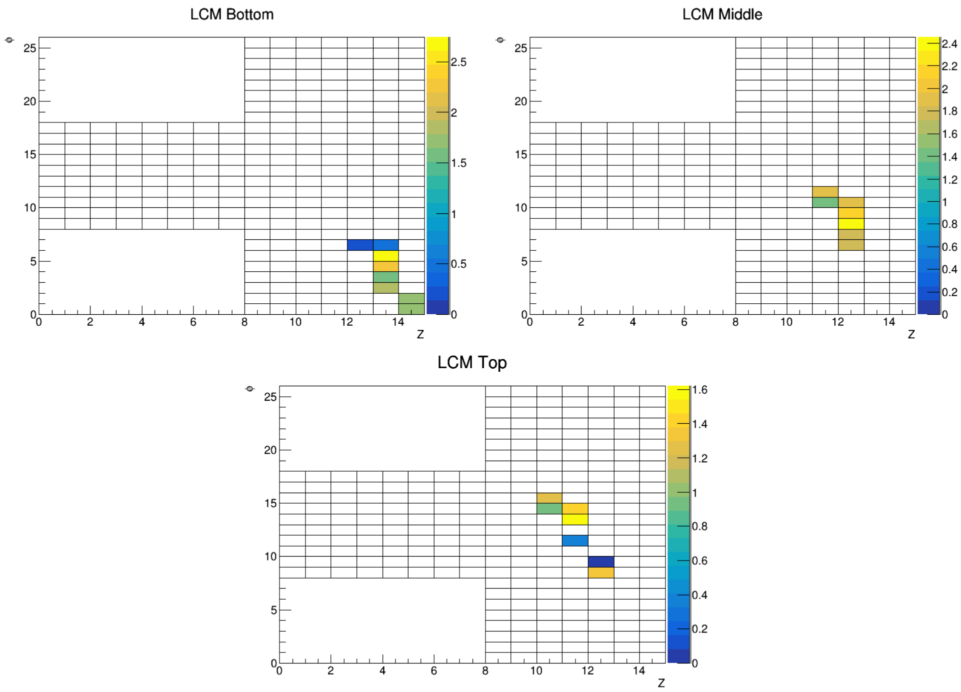


Figure 8: Example of an event display from Geant4 with the energy deposited in MeV, for the 3 LCM layers.

5 Analysis of the real data

With the simulation run, a data sample has been collected by imposing a $majority = 3$ (so at least 3 channels per board return a trigger) in the so called auto trigger mode.

A total of 13 readout boards with 60 channels each was available, translating in 12 ϕ modules per board (one ϕ module is $3LCM + 2t_0$ so one vertical unit with the same z and ϕ). The tiles are not all compact and uniform in disposition depending on the cabling, so a mapping has been performed. Also not all the channels are considered active, this because during the calibration of 2023 (done with particle beams at CERN) not all channels have been calibrated, and so they were not considered (even though they were reading correctly the signal).

As it has been told before the .root file of output is board based while for the data analysis it is preferable an event-based file. To do so I selected a proper coincidence window, which is the time for two or more boards to be considered in coincidence, and then properly reorganized the data.

Before choosing this window it is necessary to check the correctness of the data sample.

It has been noticed during the data collection that the board 4 was reporting an anomalous low gain value. The source has been identified by analyzing the channel 64, which is not connected to any SiPM and so should report only the pedestal. As it is possible to notice from figure 9 the board 4 has a broader range value compared to the others. Thus this board has been eliminated. Performing a Gaussian fit for each channel it is possible to determine the mean and σ, and from this perform a new check in which no deformities has been noticed as it is possible to see from 9.

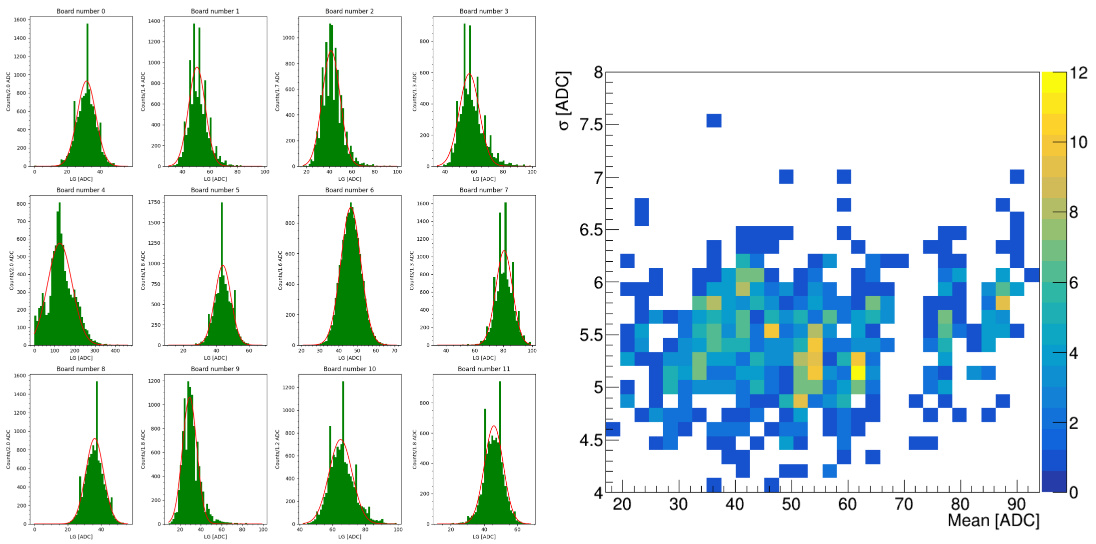


Figure 9: On the left the pedestal for all boards channel 64, with relative fit, on the left the 2D histogram of the mean and σ for all the gaussian fits (excluded board 4).

Then to select the coincidence window I need to know the trigger rate, if I select a too high value I will get multiple muon events considered as only one. An empirical estimation of the muon rate is $Rate \approx 1\ \mathrm{cm}^{-2}\ \mathrm{min}^{-1} \times$ Area of the detector ≈ 50 Hz. From the real muon rate calculated by the data we have the value 25.0 ± 0.1 Hz got by the exponential fit of the time difference between two consecutive events, as show in figure 10. From this considerations and the fact that the timestamp minimum digit is of 1 ns the coincidence window selected is 10 ns.

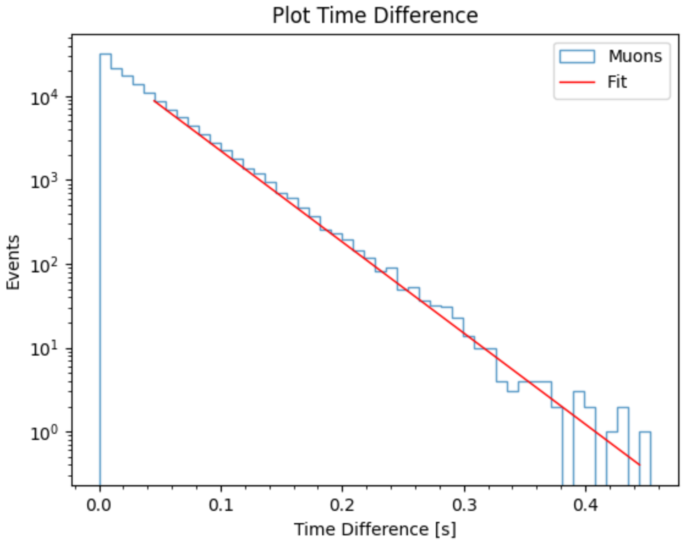
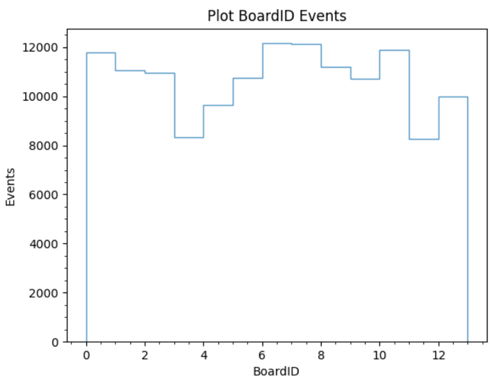


Figure 10: The histogram of the time difference between consecutive events and the proper exponential fit.

Then I checked the histogram of the events for each board (as shown in figure 11, left): as it can be seen, all boards have an almost uniform number of triggers even the anomalous board 4. Next an histogram of the multiplicity of the number of boards is done to check how the system responds (figure 11, right).



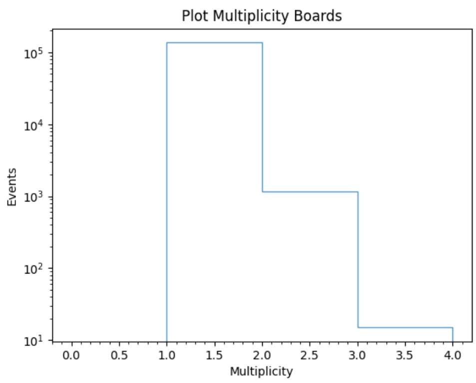


Figure 11: Number of events per board and board multiplicity.

With this decision of the window coincidence of 10 ns a proper event-based data sample can be computed. Starting from the board base data the steps necessary are

- For each channel calculate the mean and the standard deviation which should be the values of the different pedestals (necessary to account for slight deviations between each one).
- Subtract to all the channels the obtained mean value, thus becoming our reference zero, in this way there is no dependence (as we know the energy deposited is proportional to the high or low gain value from the pedestal).

- Execute the energy equalization, which is done by dividing the High Gain and Low Gain with a weight, given by the beam test of August 2023. Each weight was calculate by first doing a Landau Gaussian convoluted fit on the energy histogram, so finding the MPV. Then from this value the ratio between the MPV value minus the pedestal and the mean of all this values for all channels, gives the weight, so $\frac{MPV - pedestal}{\text{mean of the MPVs}}$.

- A variable threshold has been chosen a defined as $threshold = mean + 6 \cdot \sigma$ for the LCMs and $threshold = mean + 18 \cdot \sigma$ for the t_0. The two parameters were determined from the previous Gaussian fit. The factors of 6 and 18 were selected by comparing the histograms to minimize the pedestal; these values can be adjusted based on histogram analysis. The two threshold are different in values because the number of scintillators in the t_0 is $1/5$ and so the pedestal values are around 5 times more. This *threshold* will be called the variable threshold.

- Next I did a proper mapping in the (r, z, ϕ) position.

All this steps have been documented at [6].

In this case then I do not have an energy calibration but only an equalization. To do the energy calibration it is necessary to do a Landau Gaussian convoluted fit ([2]) of the spectrum of LG and HG with all the values for each channel taken and of the spectrum of the Geant4 simulation, of course summing over all channels to increase statistics. Figure 12 report these three fits. The calibration parameter is the ratio as $a = 0.00938 \pm 0.00011 [\frac{MeV}{ADC}]$ for the LG and $a = 0.00078 \pm 0.00002 [\frac{MeV}{ADC}]$ for the HG. The superposition of all three LG, HG (calibrated) and Geant4 spectrum is shown in figure 13.

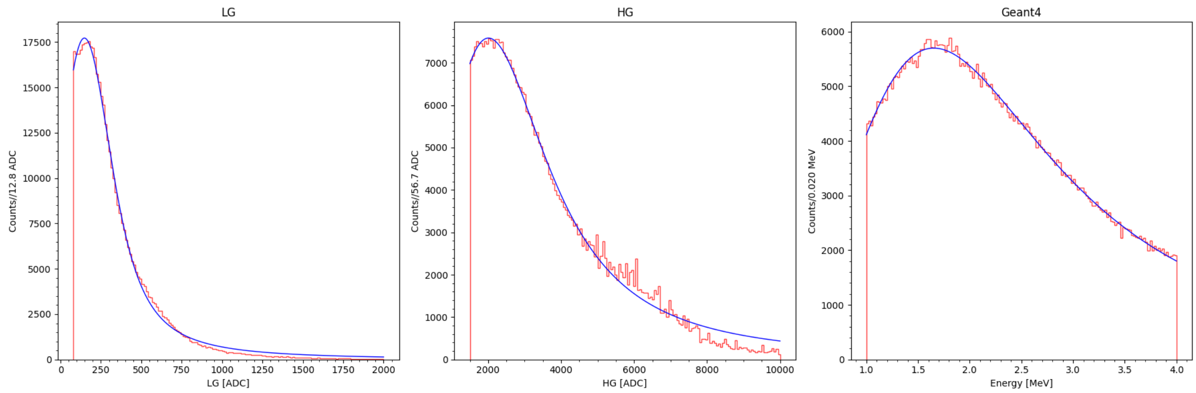


Figure 12: LG, HG and Geant4 spectrum and relative calibration fits.

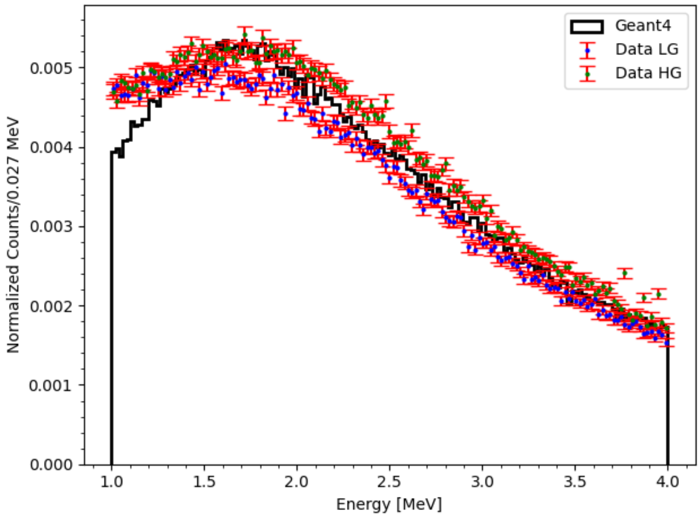


Figure 13: Energy spectrum calibrated.

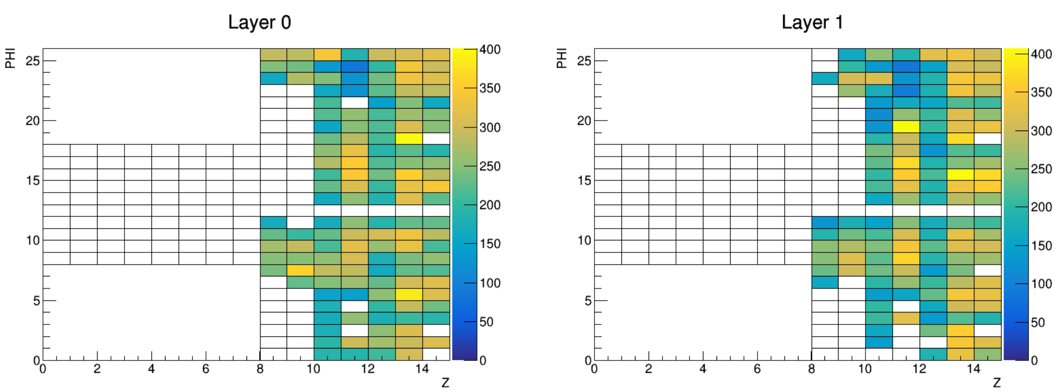
6 Uniformity

After the calibration described above the detector is efficiently calibrated and with an uniform energy response. Differences in the efficiency of each LCM or t_0 can be estimated through the ratio $D = \frac{\text{number of events - mean}}{\text{number of events + mean}}$, with mean $= \sum_{\text{ALL CHANNELS}} \frac{\text{number of events}}{\text{number of channels}}$: D is between -1 and $+1$ and it is a disuniformity parameter independent on the total number of events. But first a hit map has been done so to check the number of events. I found that the channels listed in the table 1, after the processing explained in the previous section, are marked as no more active even though they were before the formation of the event based-data set, this means that they report the pedestal value only and no MIP events. The hit maps are reported in figure 14

	1	2	3	4	5	6	7	8	9	10	11	12	13
R	0	0	0	0	1	1	1	1	1	2	2	3	4
Z	14	14	14	11	11	14	14	9	9	9	9	9	9
ϕ	0	2	18	21	0	2	18	11	22	10	22	10	10

Table 1: Dead Channels Positions.

while the disuniformity in figure 15. The same disuniformity parameter can be evaluated from the Geant4 simulation and it is between -0.1 to 0.1 similarly as for the data.



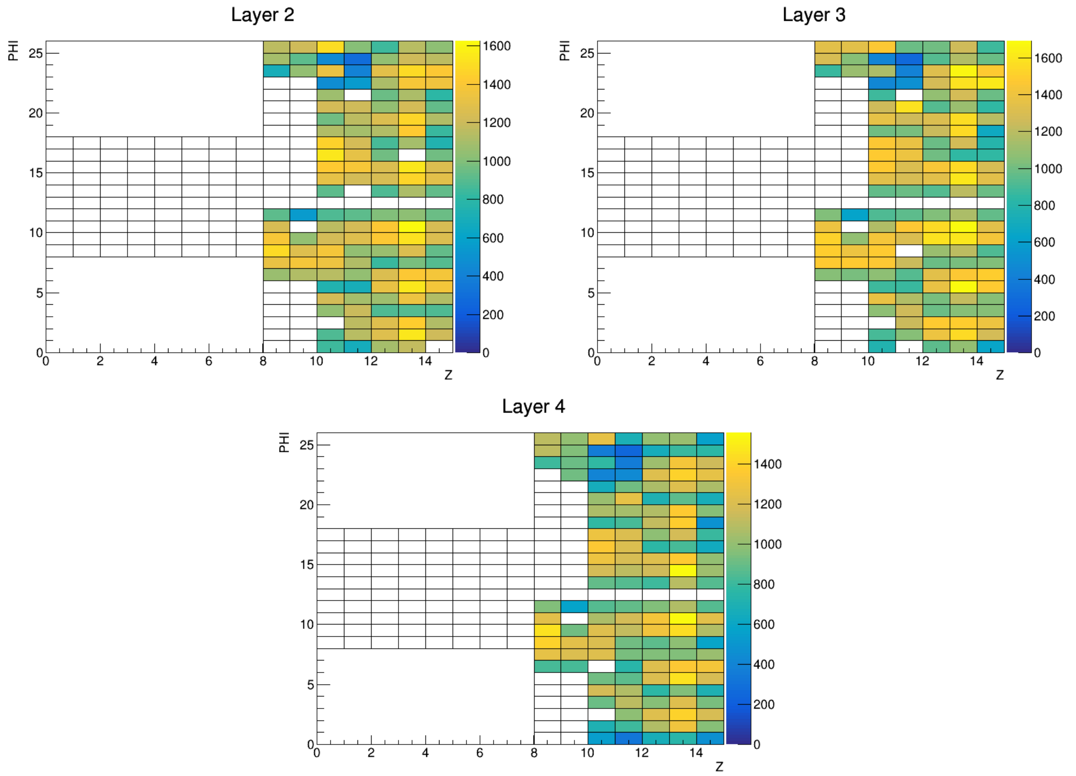
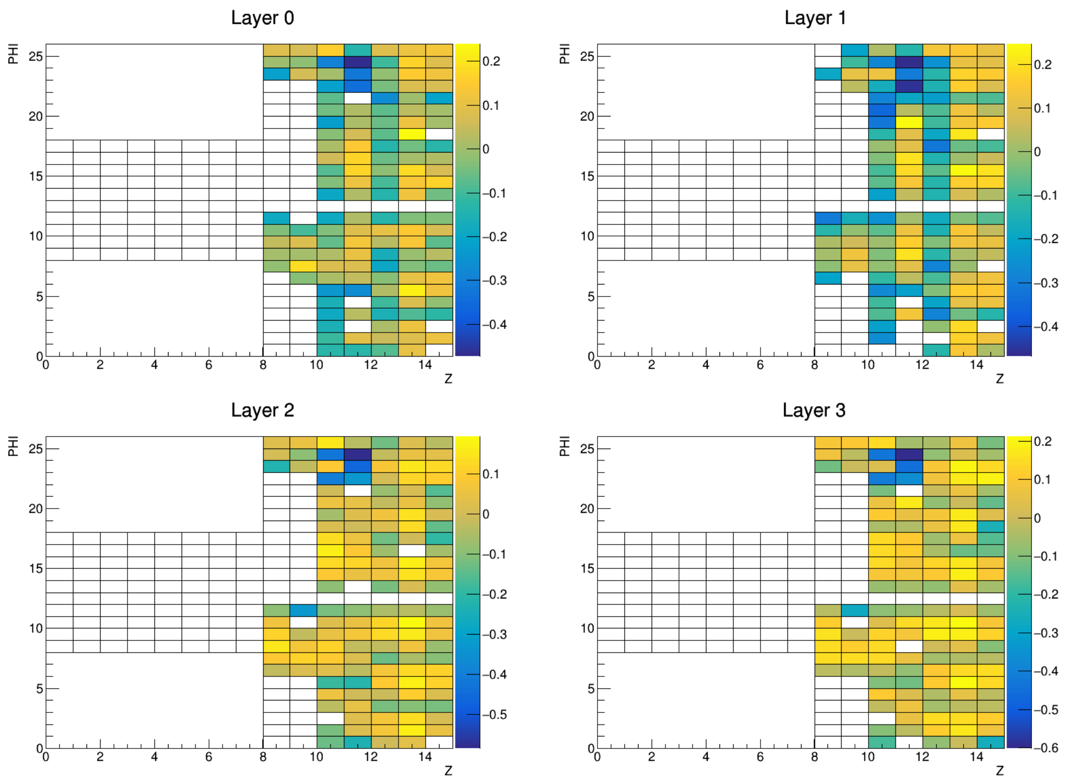


Figure 14: Hit map for the 5 layers of the data.



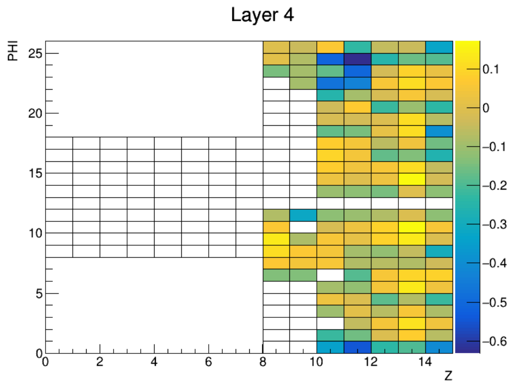


Figure 15: Disuniformity for the 5 layers of the data.

The only difference is in a region between $z = 10, 11$ and $\phi = 22, 23, 24$ for all three layers, all are connected to the same board. To further investigate this problem the LG of these channels, labeled L, and of all the other channels, labeled H, is plotted in figure 16. The sum of all the bins is normalized so it is possible to notice that the pedestal values fraction are slightly higher than expected for, and so accounting for the lower number of events above the variable threshold (so above around 80-100ADC).

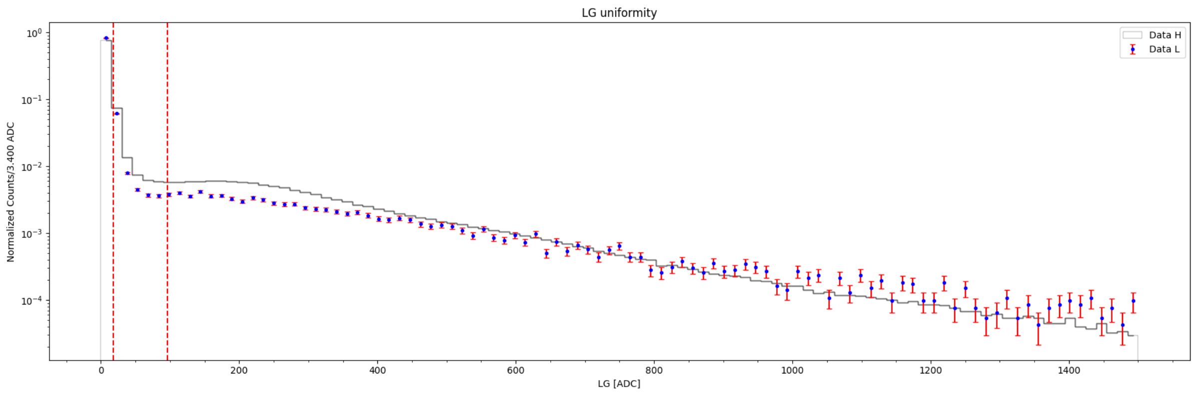


Figure 16: Low Gain for the channel in the region between $z = 10, 11$ and $\phi = 22, 23, 24$, labeled L, and all the others, labeled, H. The two red lines indicate the lowest and highest value of the variable threshold, it is possible to see that (cutting from the maximum or minimum value) the sum of the bins for L is lower than the sum for H.

It is also important to compare the triplet (r, z, ϕ) with respect to the simulation as reported in figure 17. The agreement between the simulation and the data are good for all three coordinates unless due to the disuniformity as it is possible to see from a comparison with the previous maps. It is important to point out that the histogram of r depend on the different threshold applied to the t_0.

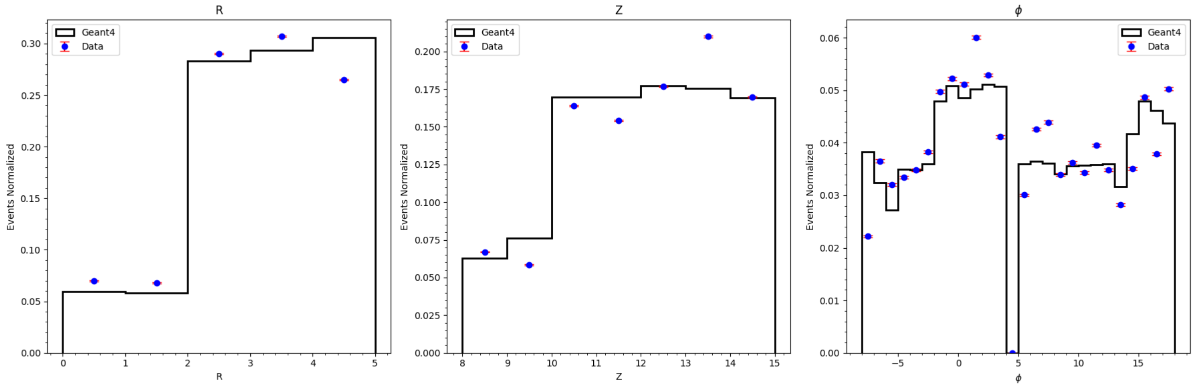


Figure 17: Position energy deposition comparison over r, z, and ϕ.

7 Event reconstruction

At this point I was able to access the quadruplet $(r, z, \phi, energy)$, using the LG as energy for the data and the Geant4 energy for the simulation (in this last case adding also the primary direction, so the real direction of the muon generated).

With this in mind I have created an event reconstruction program to fit a line defined as

$$x = 150 + t$$
$$y = p_0 + p_1 \cdot t$$
$$z = p_2 + p_3 \cdot t$$

(150 is an arbitrary choice parameter, it is only important that this is different from 0). The fit is done by minimizing the sum of $distance^2 \cdot energy$, so a program that does an energy weighted fit. We know that a three dimensional line has only four degrees of freedom and this disposition in parameters (so that $x = 150 + t$ and not $y = 150 + t$ or $z = 150 + t$) it has been chosen because of the reduced segmentation of the detector along the $z - axis$, which is in proportion to the other two directions around 4 times less segmented. (The program is documented at [6].)

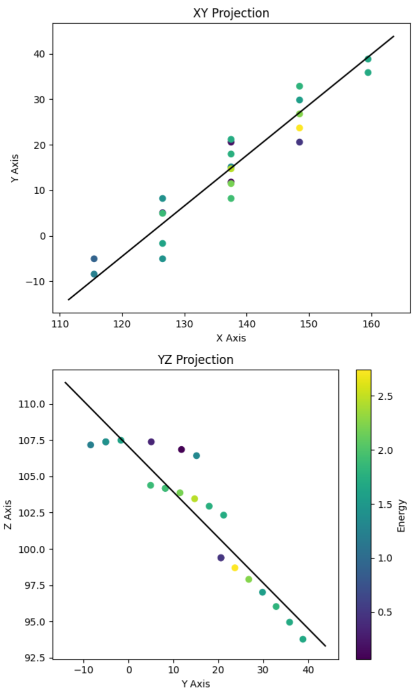


Figure 18: Example of fitted line, for the previous event display.

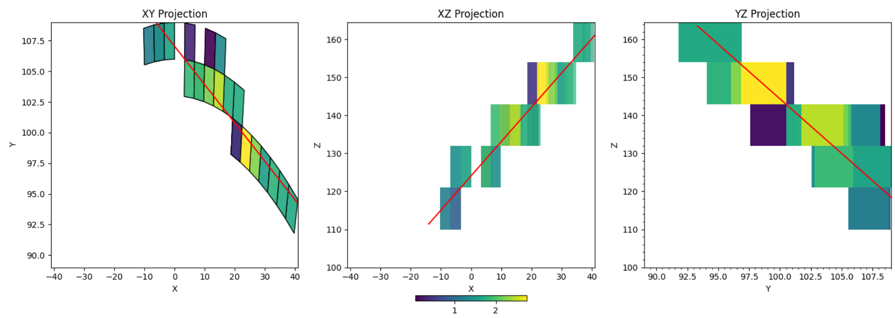


Figure 19: Example of fitted line, for the previous event display with the hit LCMs so to understand better the geometry of the event.

The azimuth angle is then defined, first by normalizing the direction vector and then as $azimuth = arctan(\frac{dirY}{dirZ}) \cdot \frac{180}{2\pi}$, so in the $y - z$ plane, while the elevation angle as $elevation = arctan\frac{dirX}{\sqrt{dirY^2+dirZ^2}} \cdot \frac{180}{2\pi}$, so in the $x - y$.

Before proceeding it is important to note that the comparison between data and initial direction it is not the correct. In particular what we do not consider in this case is the segmentation of the detector and the sampling in position of the detector. So the comparison is between the parameters of the fit from the data and the same parameters from the Geant4 events reconstructed with the same program. As seen in figure 21 there is good agreement between the data and the Geant4 results. For the elevation the agreement is lost at low angles, it has been checked what the cause could be but no clear explanation has been found. I applied three cuts

$azimuth < 85°$, $elevation > 5°$ and $NDF > 5$, this last is the Number Degree of Freedom, the number of points that are used in the fit. This cuts has been chosen because as it is possible to see from figure 20 for $azimuth > 85°$ and $elevation < 5°$ there are two anomalous peaks, due to fit failing.

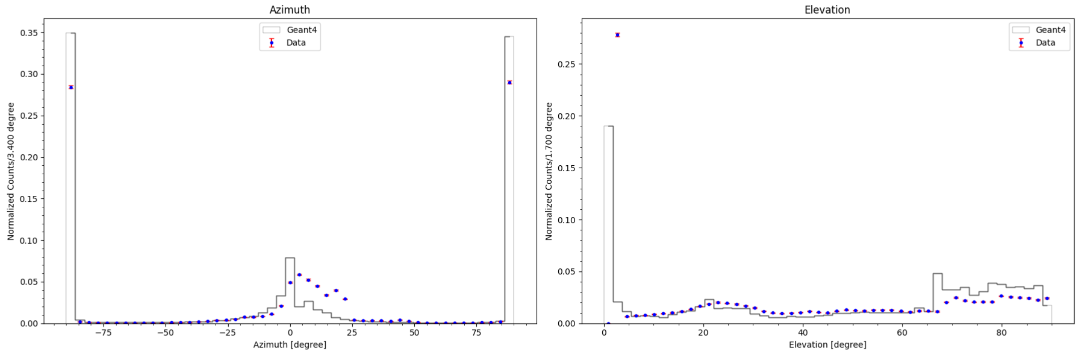


Figure 20: Angular spectrum comparison before the cuts applied.

Also the comparison with the angular spectrum does not resemble the expected one because we can have three cases

- $NDF \leq 3$ in this case the muon cross three layer in the same (z, ϕ) position, so it is not possible to determine the azimuth being a perfectly vertical line
- $4 \leq NDF \leq 7$ intermediate values in which we can determine the azimuth
- $8 \leq NDF$ good events with excellent resolution for the direction

So the goodness of the fit is dependent on the NDF and so for high elevation (low NDF) the fit does not converge or converge with worse resolution.

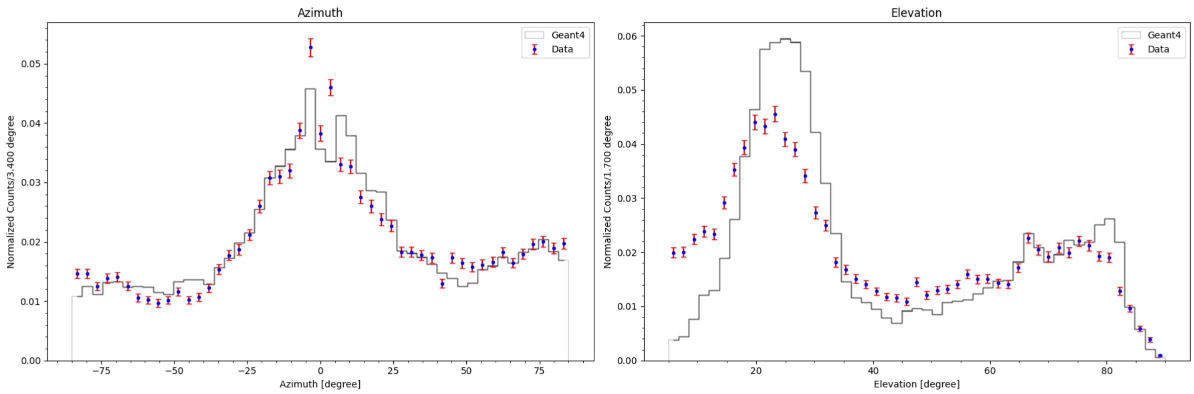


Figure 21: Data vs MonteCarlo (Geant4) with cuts $azimuth < 85°$, $elevation > 5°$ and $NDF > 5$.

To determine the resolution of the angular reconstruction, it is necessary to use the Geant4 data set and calculate the difference between the real angle and the reconstructed ones. Then it is possible to execute a Gaussian fit and the σ is the resolution. It has been done as shown in figure 23 with three cuts applied for $azimuth < 85°$,$elevation > 5°$ and $NDF > 5$. The plots in figure 22 show the bi-dimensional plots of the two variable real and reconstructed angles, it is possible to visualize all elements around the diagonal, so the correlation among reconstructed and real angles.

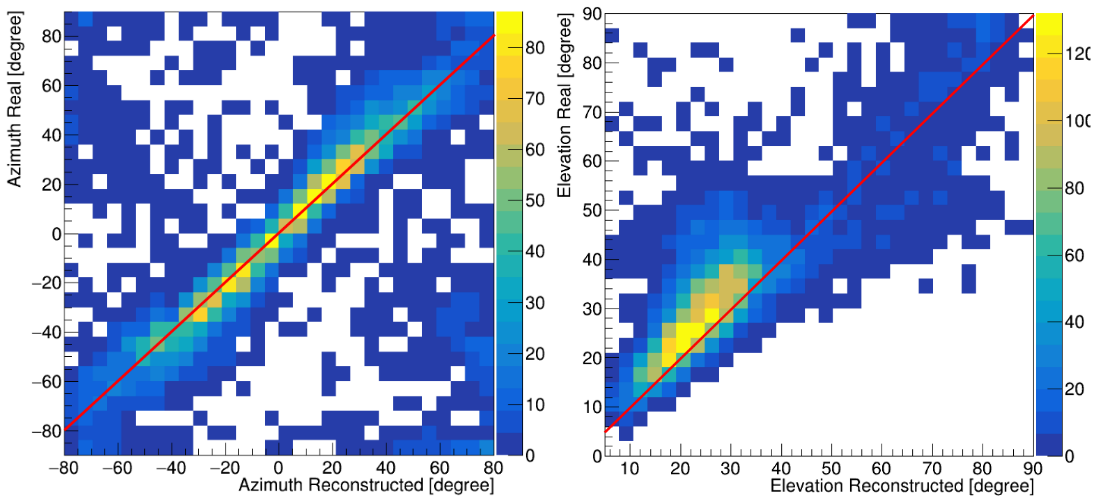


Figure 22: On the left the bi-dimensional plot of the reconstructed azimuth angle and the real one, while on the left the bi-dimensional plot of the reconstructed elevation angle and the real one.

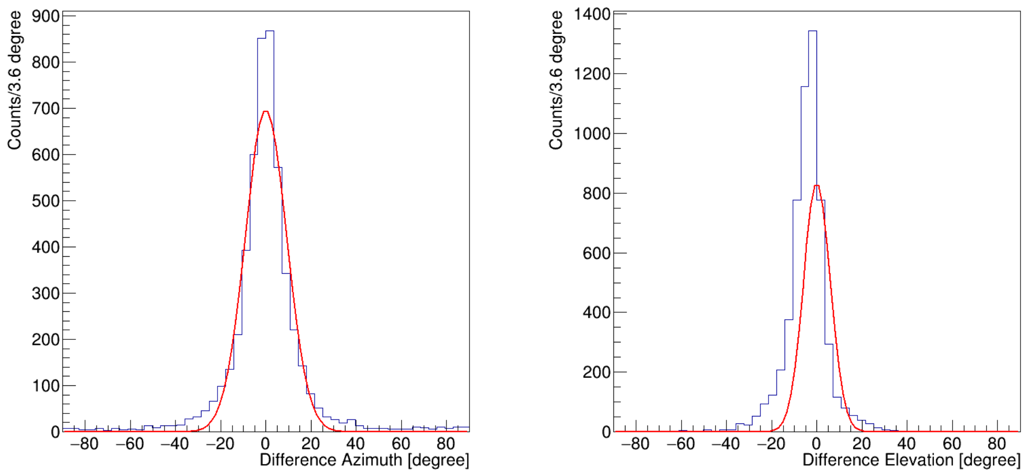


Figure 23: Angular spectrum difference for the azimuth and the elevation, with relative fit with $\mu = 0$.

The fit results return a resolution of $9.3 \pm 0.1°$ for the azimuth and of $6.0 \pm 0.2°$ for the elevation. The fit has been performed with a Gaussian with $\mu = 0$, as it is possible to see an offset of around $5°$ is present in the elevation. The same fit has been executed with a free μ getting for the elevation $\mu = -3.9 \pm 0.1°$ and for the azimuth $\mu = -0.2 \pm 0.2°$, the second is compatible with zero while the first one it isn't; the resolutions are respectively $8.6 \pm 0.2°$ and $5.1 \pm 0.1°$.

It is possible to do the same procedure for the events divided for different NDF, selecting events with only one specific NDF. As it is shown in the figure 24 if we select lower NDF we get a worse resolution for both azimuth and elevation.

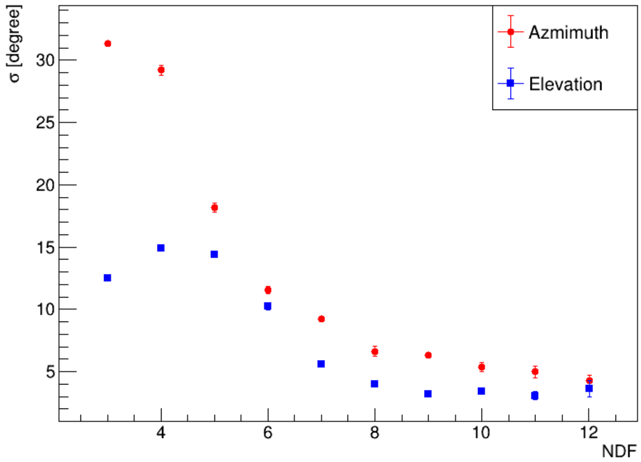


Figure 24: Resolution as a function on the NDF.

8 Conclusion

In this document I reported the results on three main items:

- a method to filter data and obtain a good sample
- the disuniformity of the detector
- the event reconstruction analysis

The disuniformity point toward anomalies in the response on the channels between $z = 10, 11$ and $\phi = 22, 23, 24$ and further analysis is necessary. As it is important to correct the dead channels perhaps due to some physical problem and not computational.

For all the comparisons with the Geant4 simulation good agreement has been found but improvement on the elevation calculation can solve the present anomaly. The resolution calculated is satisfactory and fully determined by the available segmentation.

References

[1] F. Acerbi et al. *Design and performance of the ENUBET monitored neutrino beam*. 2023. arXiv: 2308.09402 [hep-ex]. URL: https://arxiv.org/abs/2308.09402.

[2] The Landaupy Developers. *A pure python implementation of the Landau distribution*. Accessed: 2024-07-20. 2024. URL: https://pypi.org/project/landaupy/.

[3] ENUBET Collaboration. *ENUBET: Enhanced NeUtrino BEams from kaon Tagging*. https://www.pd.infn.it/eng/enubet/. Accessed: 2024-10-03. 2024.

[4] Particle Data Group. *Cosmic Rays*. Accessed: 2024-07-20. 2022. URL: https://pdg.lbl.gov/2022/reviews/rpp2022-rev-cosmic-rays.pdf.

[5] A. Longhin and F. Terranova. *Enhanced NeUtrino BEams from kaon Tagging (ENUBET)*. 2022. arXiv: 2203.08319 [hep-ex]. URL: https://arxiv.org/abs/2203.08319.

[6] Davide Marchesini. *ENUBET Project Repository*. GitHub repository. 2024. URL: https://github.com/DavideMarchesini/ENUBET.

[7] Nicola Zurlo et al. "A new Monte Carlo muon generator for cosmic-ray muon applications". In: *PoS* CompTools2021 (2022), p. 019. DOI: 10.22323/1.409.0019.